Assignment 2 – LLM Document Analysis

Extracting new product announcements from SEC 8-K filings

Student: Hegel Gonzalez **Date:** November 4, 2026

GitHub: https://github.com/hegel24horas/sec-8k-product-extractor

1. Objective

The goal is to read SEC Form 8-K filings from S&P 500 companies and use a large language model (LLM) to extract newly announced products into a structured CSV with five fields: company name, stock ticker, filing time, new product and a product description of fewer than 180 characters.

2. Data collection

All data comes from the public SEC EDGAR APIs, requested with a descriptive User-Agent and throttled to under 10 requests per second as SEC fair-access rules require.

- **Tickers:** the S&P 500 constituent list (Wikipedia), shuffled with a fixed seed (42) so the sample spans many industries and is reproducible.
- **Ticker to CIK:** files/company_tickers.json.
- **CIK to filings:** data.sec.gov submissions API, which gives the filing form, items and acceptance timestamp. I keep only 8-Ks that report Item 7.01 (Regulation FD) or Item 8.01 (Other Events), the items companies use for press releases, and take up to two per company.
- **Text:** the main 8-K document plus its EX-99 exhibits, parsed with BeautifulSoup. This matters because the product announcement is almost always in the press-release exhibit, not in the short cover document.

The run stopped after 100 filings, the cap suggested in the assignment.

3. Methodology

Step 1 – NER preprocessing (spaCy). Each filing is split into sentences with spaCy's en_core_web_sm model. Sentences are kept if they contain a PRODUCT entity or launch language (launch, introduce, unveil, announce, available, new). Up to about 7,000 characters are kept, which removes legal boilerplate and cuts LLM cost. The ORG and PRODUCT entities found are passed to the LLM as hints.

Step 2 – LLM extraction. The excerpt is sent to OpenAI gpt-4o-mini (temperature 0, JSON mode). The prompt asks only for new products, services or platforms, states that most filings have zero or one, and explicitly excludes financial instruments (notes, bonds, stock), earnings, presentations, acquisitions, and, for pharma, trial-data updates on already known drugs. The model returns a product name and a one-sentence description.

Step 3 – Validation and output. Company name and filing time come from SEC metadata rather than the model, so they are exact. Descriptions are cut at a word boundary to at most 179 characters. A keyword filter removes any remaining financial-instrument results. Rows are written to products.csv with all fields quoted.

I used the OpenAI API instead of a local Ollama model because Ollama was not installed on my machine; the code is a single function, so a local model could replace it.

4. Results

From 100 filings the pipeline produced 10 rows covering 6 companies:

- Hasbro: two new digital games (Exodus, Warlock).
- Altria: two new flavors in the PLUS portfolio (Blueberry Mint, Mango Pineapple).
- Bristol Myers Squibb: Zenbexus, newly approved, and mezigdomide.
- Gilead (anito-cel CAR T therapy), Arista Networks (AI networking capabilities) and Becton Dickinson (ChloraPrep).

Only about 10% of 8-Ks announce a product. Most Items 7.01 and 8.01 cover financings, investor presentations, leadership changes or dividends, so a low yield is expected rather than a failure.

5. Limitations and lessons learned

- **Precision:** a manual review found two doubtful rows. ChloraPrep is a shipment resumption, not a new product, and Camzyos is an existing drug with a new regulatory filing. The LLM sometimes treats an announcement about a product as a launch.
- **Recall:** I did not hand-label a ground truth, so recall is unmeasured. Items such as 2.02 (earnings releases) were excluded and may hide some launches.
- **Iteration:** a first version returned bonds and preferred stock as “products” and read only the cover document. Reading the exhibits, tightening the prompt and adding the filter fixed most of this.
- **Sampling:** the 100-filing sample is small and random, so results are not representative of the full index.

6. How to reproduce

Create a .env file with OPENAI_API_KEY and SEC_USER_AGENT, run pip install -r requirements.txt and python -m spacy download en_core_web_sm, then python sec_8k_product_extractor.py. The script writes products.csv.